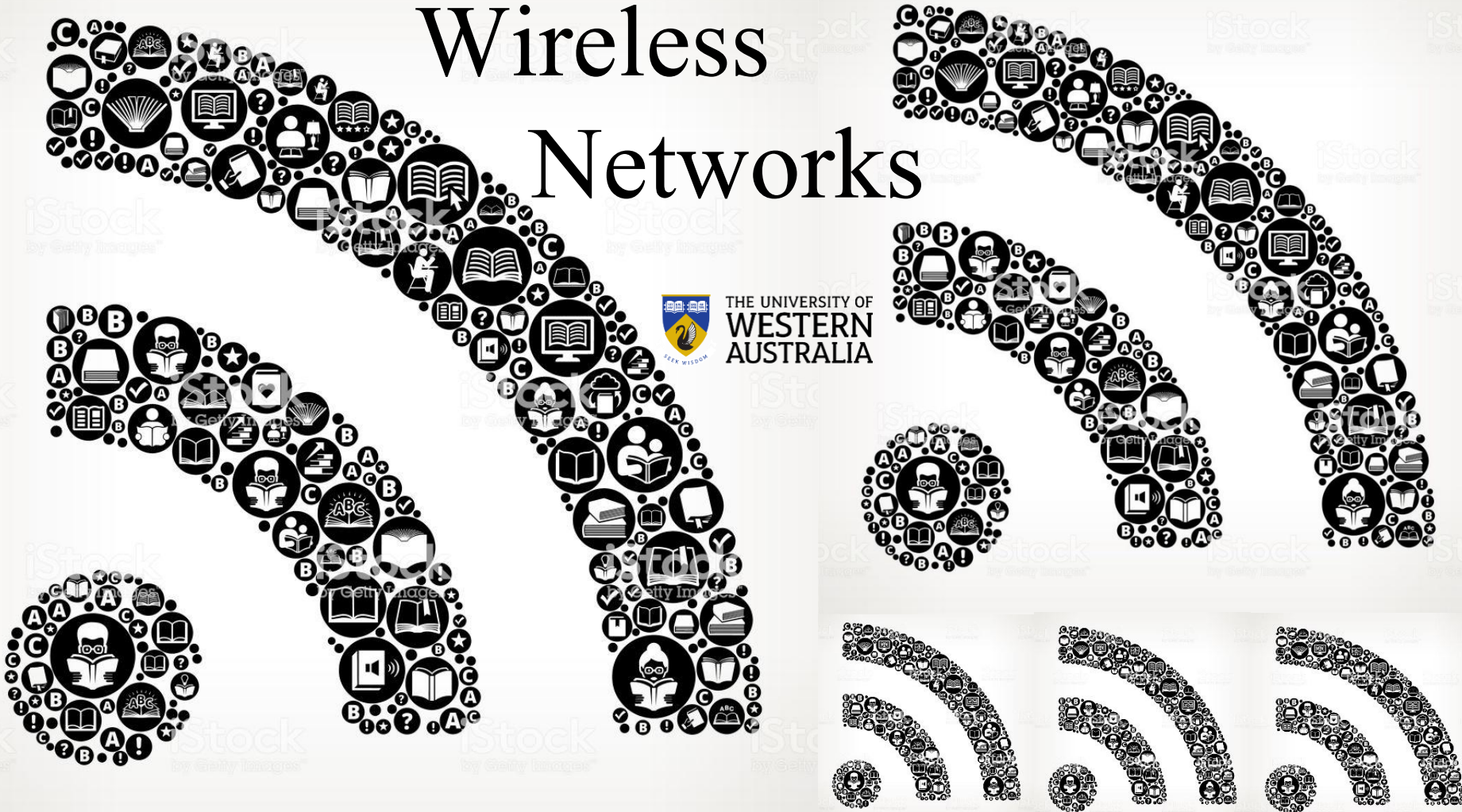


Wireless Networks



THE UNIVERSITY OF
WESTERN
AUSTRALIA



Outline

- ☐ Wireless LAN
- ☐ MACA
- ☐ CSMA/CA
- ☐ IEEE 802.11
- ☐ Network Layer

- ❑ Wireless LAN (**WLAN**) is standardized by IEEE
 - The main wireless LAN standard is **802.11** (Wi-Fi)
 - Wi-Fi was introduced in 1997
 - Since its first adoption, new versions were released
 - ❖ Versions include a, b, g, n, h, i, ac, ad, af, ah, ai, aj, aq, ax, ay and so forth
 - The current Wi-Fi generation is called **Wi-Fi 7** (also known as IEEE 802.11**be**)
 - ❖ Maximum link rate of 46 Gbps

❑ 802.11 networks can be used in two **modes**:

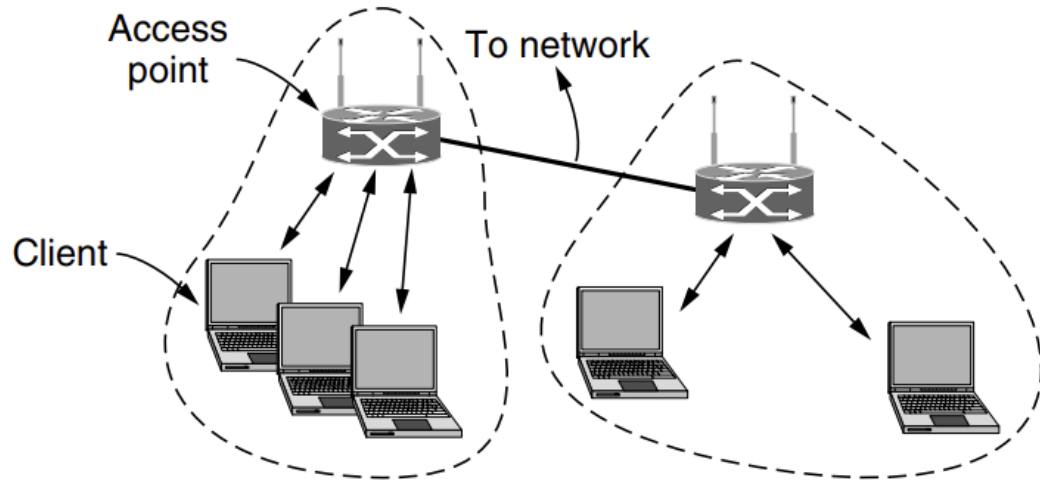
➤ **Infrastructure** mode:

- ❖ The most popular mode is to **connect clients to another network**, such as
 - Internet
 - Enterprise networks (e.g., intranet)
- ❖ Each client is associated with an **Access Point (AP)**
- ❖ Clients can send frames to other clients **via their APs**

➤ **Ad-hoc** mode (peer-to-peer):

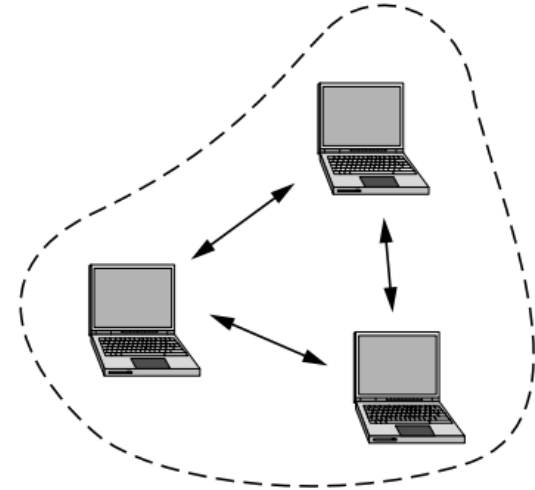
- ❖ This mode is a collection of computers that are **associated** so that they can **directly** send frames to each other
- ❖ There is **no** access point

Wireless LAN



(a)

Infrastructure mode



(b)

Ad-hoc mode

Wired vs. Wireless LAN

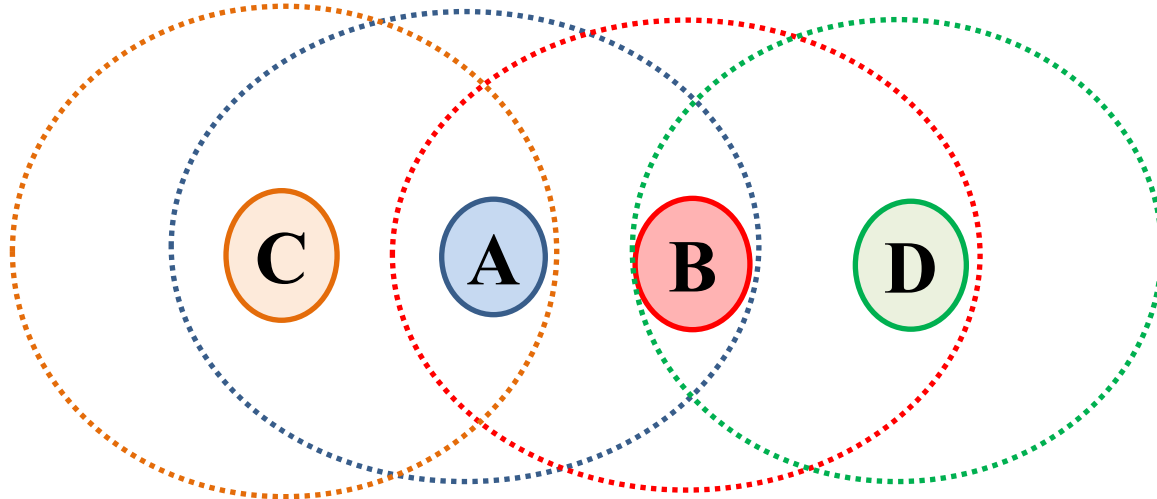
- ☐ Wireless systems **cannot normally detect a collision** while it is **occurring**.
- ☐ The received signal at a station may be tiny, perhaps a **million times fainter** than the signal that is being transmitted.
- ☐ Finding it is like looking for a ripple on the ocean.
- ☐ **Acknowledgements** are used to **discover transmission failure**

Wired vs. Wireless LAN

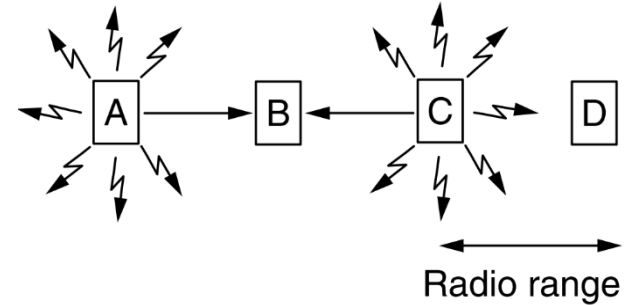
- ❑ A station on a **wireless LAN** may not be able to transmit frames to or receive frames from **all** other stations
 - Stations have a **limited** radio range

- ❑ In **wired LANs**, when one station sends a frame, **all** other stations receive it
 - The **absence** of this property in wireless LANs causes a variety of complications

- ❑ A **naive** approach to using a wireless LAN might be to try **CSMA**
 - Just listen for other transmissions and only transmit if **no one else** is doing so
 - This protocol **is not really a good way** to think about wireless
 - ❑ What matters for reception is **interference at the receiver**, **not** at the sender

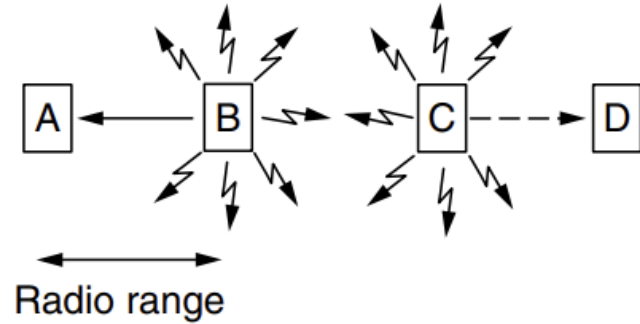


Hidden Terminal Problem



- ❑ What happens when A and C transmit to B?
 - If A sends and then C immediately senses the medium, it will not hear A because A is placed **out of the range of C**.
 - Thus, C will **falsely** conclude that it can transmit to B.
 - If C does start transmitting, it will **interfere** at B, wiping out the frame from A.
- ❑ The problem of a station not being able to detect a potential competitor for the medium because the **competitor is too far away**.

Exposed Terminal Problem

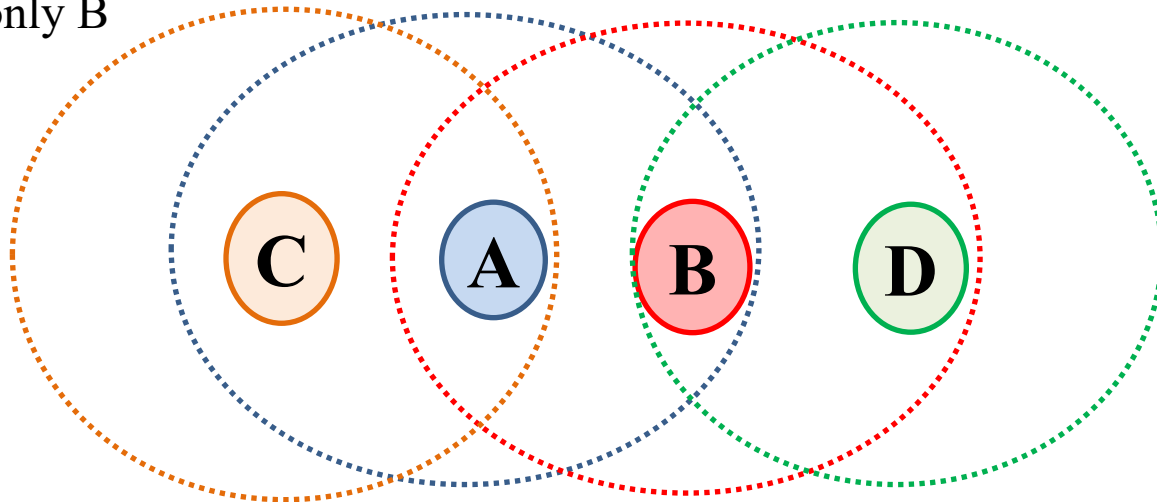


- ❑ B transmitting to A at the same time that C wants to transmit to D
- ❑ If C senses the medium, it will hear a transmission and **falsely** conclude that **it may not send to D**
- ❑ We want a MAC protocol that prevents this kind of deferral from happening because it **wastes bandwidth**

- ❑ The challenge is that, before transmitting, a station needs to know whether there is any **radio activity** near the **receiver**
 - CSMA **only** indicates whether there is activity near the transmitter by **sensing the carrier signal**
- ❑ In a **wired** network, **all signals** propagate to **all stations**
 - Typically, only **one transmission** can occur at a time
- ❑ In short-range **wireless** systems, **multiple** transmissions can occur **simultaneously**
 - This is possible if the transmissions are intended for **different destinations** that are **out of each other's range**

- ❑ **MACA** (Multiple Access with Collision Avoidance) is an early protocol that helps solve these problems in wireless LANs
- ❑ The sender makes the **receiver send a short frame** so nearby stations know the channel is busy and avoid transmitting while the main data frame is sent
- ❑ IEEE 802.11 reduces collisions using a protocol known as **CSMA/CA** (Carrier Sense Multiple Access with Collision Avoidance)

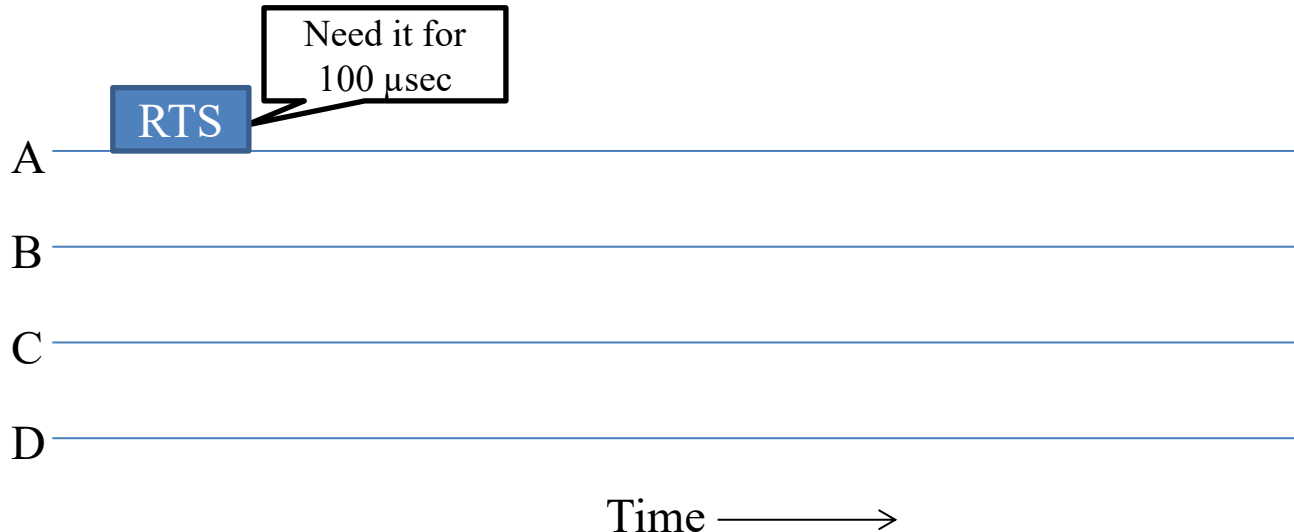
- ❑ Consider the situation of 4 nodes. You can imagine that we have 4 nodes left-to-right, named C, A, B, D.
 - A wishes to communicate with B
 - C can hear only A
 - D can hear only B



- ❑ A observes an idle medium, and initially sends a *Request to Send* (RTS) frame to B



- ❑ This frame includes a field indicating how long (in microseconds) the actual data frame will be
 - i.e. how long the sender wishes to *hold* the medium.

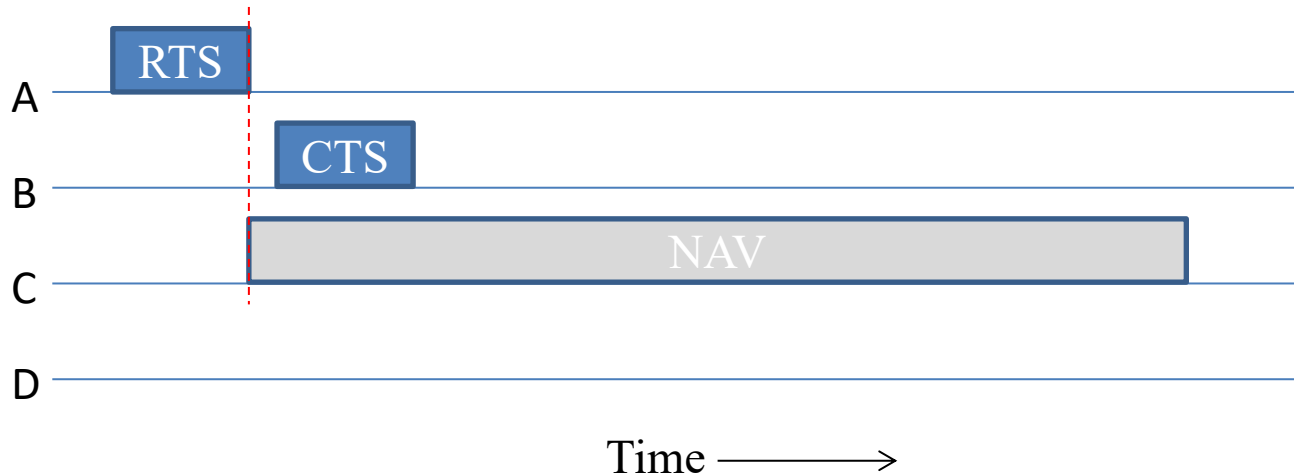


- ❑ When B receives the RTS it replies with a *Clear to Send (CTS)* frame, also carrying the length of the data frame.

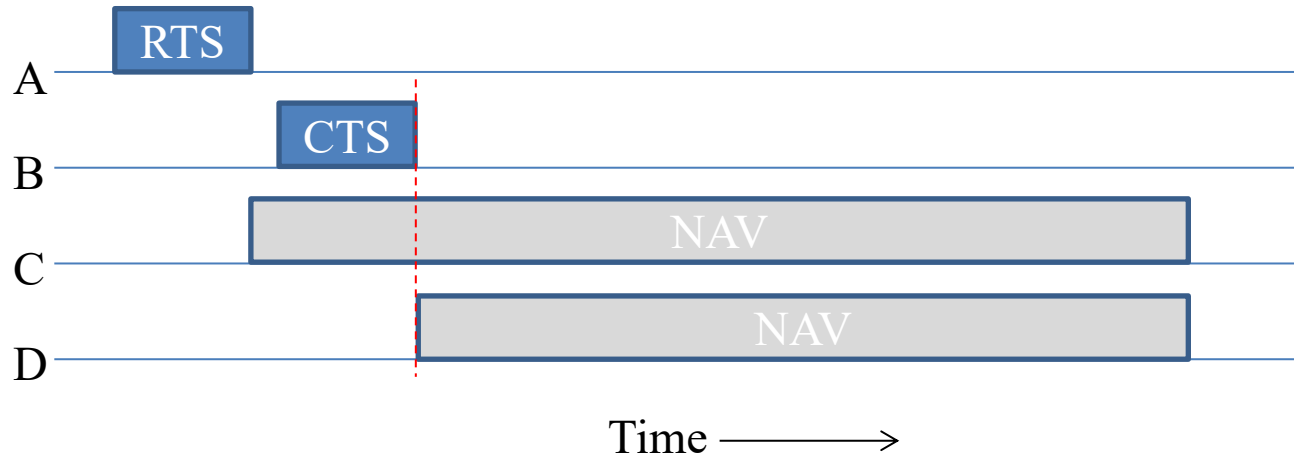


- ❑ 802.11 defines channel sensing as a combination of **physical** sensing and **virtual sensing**
 - Physical sensing checks the medium for the **presence** of a **valid signal**.
- ❑ With virtual sensing, stations keep a **logical record** of channel occupancy using the **NAV** (Network Allocation Vector)
- ❑ Each frame has a **NAV field** that shows **how long** the transmission will take
- ❑ Stations that overhear **NAV** understand that the channel will remain **busy** for the duration specified by the NAV, **regardless** of whether they can **physically** sense a signal

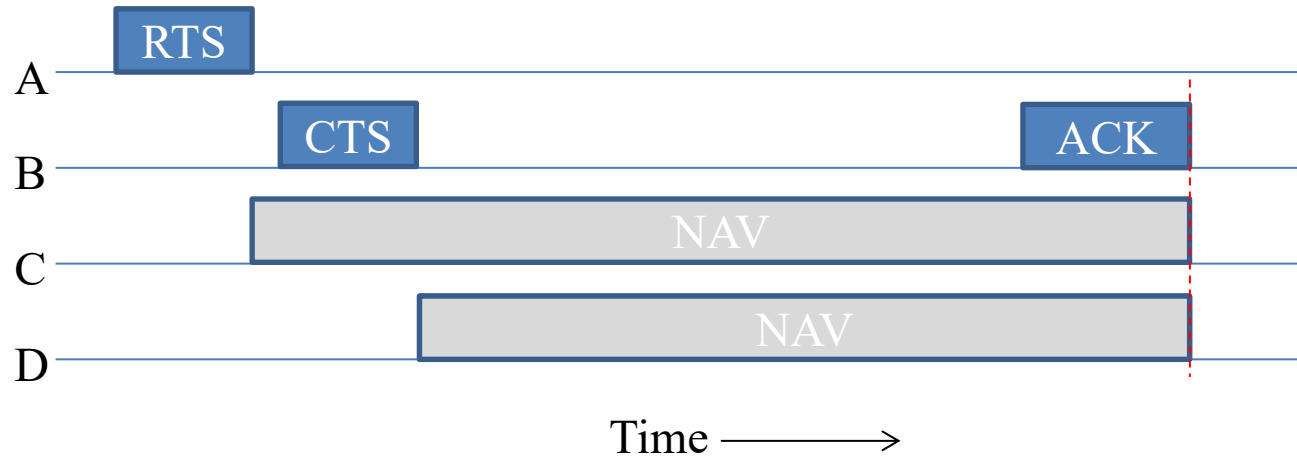
- ❑ Any other node hearing the RTS frame (e.g. **C**) knows that **A** is making a request, and should not transmit until the indicated length/time has elapsed.



- ❑ Any node hearing the CTS frame (e.g. **D**) must be close to the receiver (**B**) and therefore should also not transmit for the indicated length of time.



- ❑ When the receiver (B) successfully receives a data frame, it must reply with an **ACK** frame. All other nodes must wait for this ACK frame before transmitting.



- ❑ CSMA/CA, combined with physical and virtual sensing, forms the **core of the 802.11 protocol**
- ❑ However, **additional mechanisms** have been developed to support its operation, each addressing practical requirements
 - We will look at them briefly

- ❑ The primary strategy for improving transmission success is to **reduce the transmission rate**.
- ❑ Lower rates use **more robust modulation schemes**, which are **more likely** to be **correctly received** at a given signal-to-noise ratio.
- ❑ When frame loss is **high**, a station **reduces** its rate.
- ❑ When losses are **low**, it may occasionally try a **higher rate** to assess whether it can be sustained.

Sending shorter frames

- ❑ Another strategy to increase the likelihood of successful transmission is to use shorter frames.
 - If the probability of a bit error is p , the probability that an n -bit frame is received correctly is $(1 - p)^n$
 - For example, when $p = 10^{-4}$, the probability of correctly receiving a full Ethernet frame (12144 bits) is less than 30%
 - If the frame length is reduced to one-third (4048 bits), about two-thirds of the frames are received correctly

Sending shorter frames

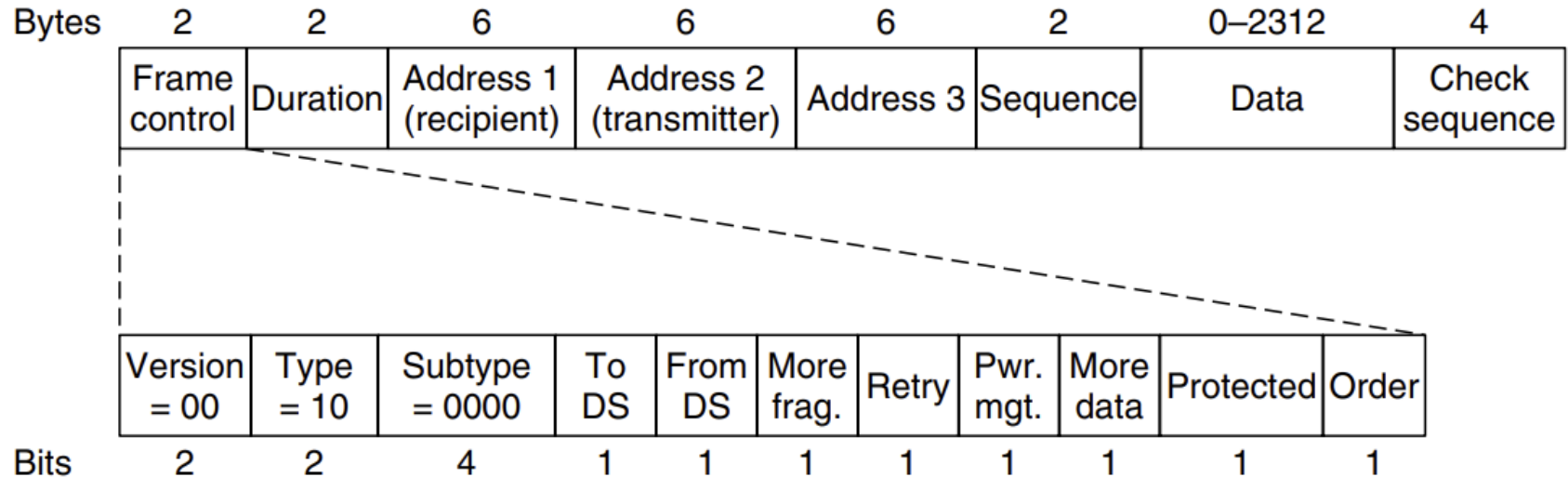
- ❑ Frame size can be reduced by
 - Limiting the **maximum message size** from the network layer or
 - By using **fragmentation**
 - ❖ Frames may be split into **smaller fragments (IEEE 802.11)**
- ❑ The fragment size is **not fixed** by the standard and can be adjusted

IEEE 802.11 Frame Structure

- ❑ The 802.11 standard defines three types of frames:
 - Data
 - Control
 - Management

- ❑ Each frame type includes a header with fields used by the MAC sublayer

IEEE 802.11 Data Frame Structure



IEEE 802.11 Services

- ❑ The 802.11 standard defines the services required for **clients**, **access points**, and the network connecting them to form a compliant wireless LAN.

- ❑ These services are grouped into several categories including:
 - Association
 - Reassociation
 - Authentication
 - Distribution
 - Privacy

Association Services

- ❑ The association service allows mobile stations to connect to **access points** (APs).
- ❑ This service is typically used when a station comes within the **radio range** of an AP. At that point, the station discovers the **AP's identity** and **capabilities**.
- ❑ The station sends a request to associate with the AP.
 - The AP may accept or reject the request
- ❑ These **capabilities** include **supported data rates**, **security features**, **power-saving options**, quality of service support, and more.

Reassociation Services

- ❑ Reassociation enables a station to **change its preferred AP**, which is essential for mobile devices transitioning between access points within **the same extended 802.11 network**.
- ❑ If performed correctly, **no data** will be lost while **moving between access points**.
- ❑ 802.11, like Ethernet, is just a **best-effort** service.
- ❑ Either the station or the AP may also **disassociate**, breaking their relationship.

Authentication Services

- ❑ Stations must **authenticate** before **transmitting frames via the AP**. The authentication process varies depending on the selected security scheme.
- ❑ **WPA2** (Wi-Fi Protected Access 2), based on the **802.11i** standard, is the recommended **security scheme**.
- ❑ With WPA2, the AP may communicate with an authentication server that maintains a **username and password** database to verify whether a station is authorized to access the network.
- ❑ Alternatively, a **pre-shared key** (i.e., a network password) can be used.

- ❑ Once frames reach the AP, the **distribution service** determines how they are **routed**.
- ❑ If the destination is connected to **the same AP**, the frames are transmitted directly over the wireless medium.
 - Otherwise, they are forwarded through the **wired network**
- ❑ The **integration service** handles any necessary **translation for frames** leaving or entering the 802.11 LAN, such as when connecting the wireless network to the Internet.

Privacy Services

- ❑ To ensure **confidentiality** in a wireless LAN, transmitted data must be **encrypted**.
- ❑ This is achieved through a privacy service that handles **encryption** and **decryption**.
- ❑ In WPA2, encryption is based on the **AES** (Advanced Encryption Standard), a U.S. government standard approved in 2002.
- ❑ The **encryption keys** are established during the **authentication** process.

Network Layer



- ❑ The Data Link Layer had the responsibility of **reliably** transmitting *frames* across a single wire (or wireless) link.
- ❑ The Network Layer's responsibility is to get *packets* from an actual **source** machine to a **destination** machine.
 - There may be many hops along the way.
- ❑ The **Network Layer** is thus the **lowest OSI Layer** that has to deal with *end-to-end* transmission.

Introduction

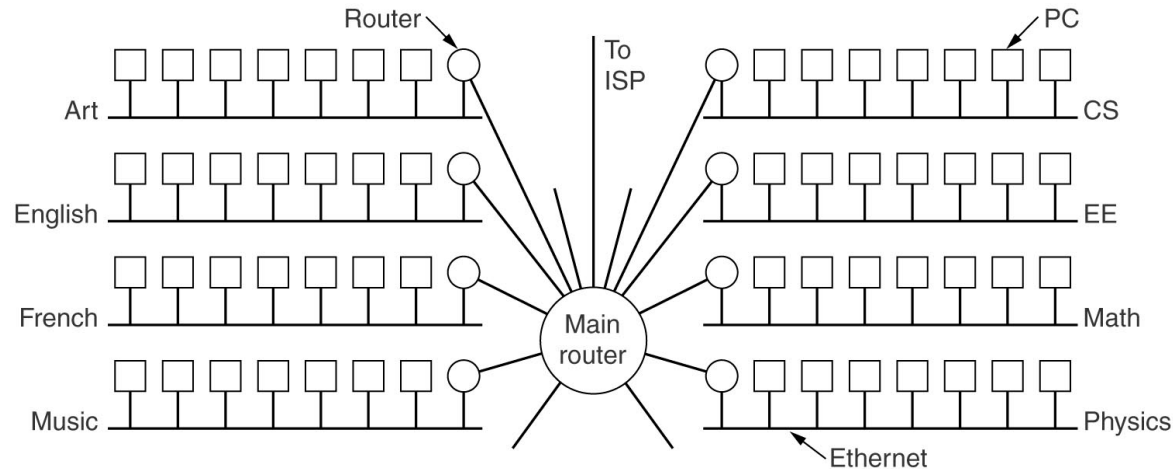
- ❑ The **Network Layer** must be aware of **the immediate topology** of its subnet (its neighbours) to make a routing decision that progresses toward the destination
- ❑ It also tries to avoid introducing **congestion on links**, and avoid *already* congested links.
- ❑ If the source and destination machine are on different networks, we must support *internetworking*.

Network Layer Design Objectives

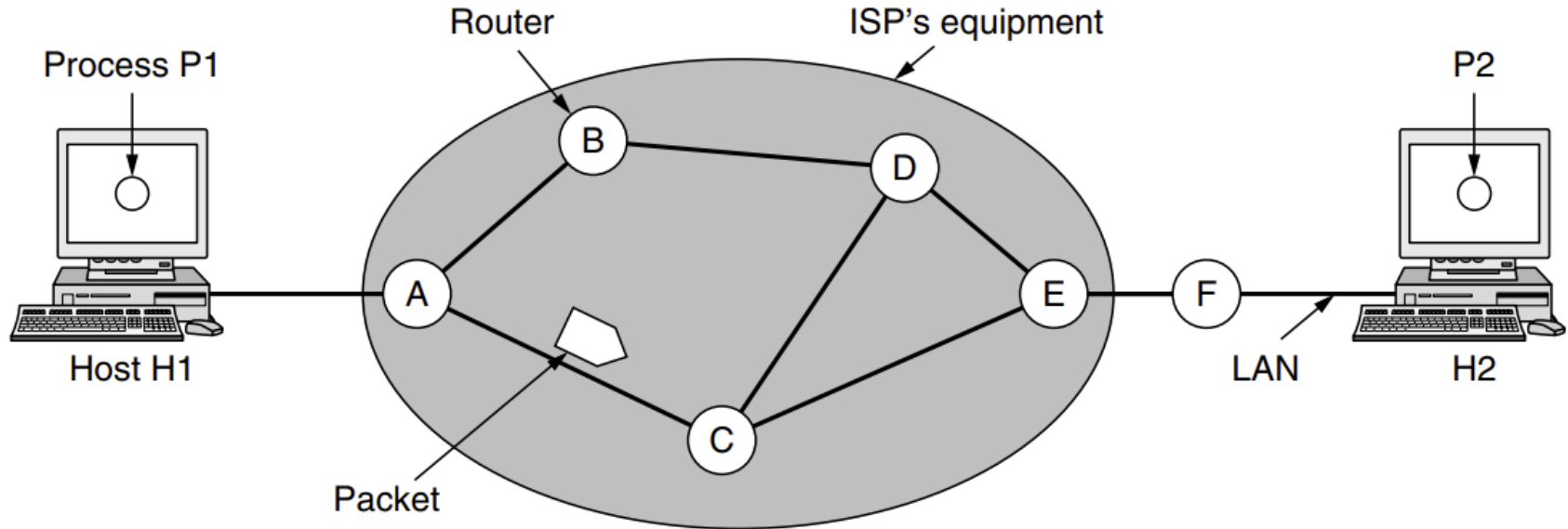
- ❑ The design objectives of an effective Network Layer are:
 - to be **independent** of processor/**communication** technology
 - to be **independent** of the number, type and **topology** of the subnets
 - to provide a **uniform addressing** scheme for **all hosts** in the network

Hosts and the Subnet

- ❑ The **Network Layer** software **runs** in dedicated devices termed **routers**
- ❑ An ISP (Internet Service Provider) is **a network of routers** and links that connects **end hosts to the wider Internet** and forwards packets **between different networks**



Store-and-Forward Packet Switching



Store-and-Forward Packet Switching

- ❑ When a host sends a packet, it is first transmitted to the **nearest router**, either within the local network or via a **point-to-point** link to the **ISP**
- ❑ The router stores the packet until it has been **fully received** and verified using a **checksum**
- ❑ It then forwards the packet to the **next router** along the path
- ❑ This process continues until the **packet reaches its destination**
- ❑ This approach is known as **store-and-forward** packet switching

- ❑ The primary function of the network layer is to **route packets** from the source to the destination
- ❑ In most networks, this process involves **multiple hops**
- ❑ The design of routing algorithms and the data structures they rely on is a **key aspect of the network layer**
- ❑ The routing algorithm is the component of the network layer responsible for determining **which output link** an incoming packet should be **forwarded to**

- ❑ A router can be viewed as consisting of two main processes:
 - **Forwarding**: handles incoming packets by determining the appropriate **outgoing link** using the routing table
 - **Routing**: Constructing and updating the routing **tables**

- ❑ Routing algorithms are generally classified into two main types:
 - **Nonadaptive (static)** algorithms: make routing decisions without considering current network conditions, such as changes in topology or traffic levels
 - ❖ Routes between all pairs of nodes are precomputed **offline** and loaded into routers when **the network starts**.
 - ❖ They are best suited for simple and stable network environments
 - **Adaptive (dynamic)** algorithms: adjust their routing decisions in response to changes in network topology and, in some cases, traffic conditions.

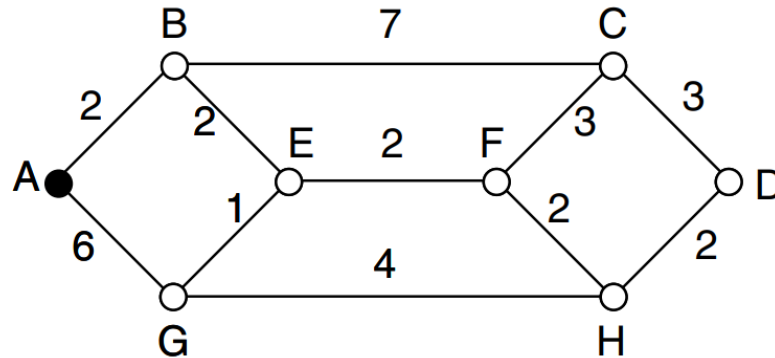
- ❑ **Adaptive** routing algorithms vary based on:
 - **Source of information**: whether data is obtained locally, from neighboring routers, or from the entire network
 - **Update timing**: whether routes are updated only when topology **changes** occur or **periodically** (e.g., every ΔT seconds) as network load varies
 - **Optimization metric**: such as shortest distance, minimum hop count, or estimated transmission delay

- ❑ The idea is to represent the network as a **graph**, where
 - Each **node** corresponds to a **router**
 - Each **edge** represents a **communication link**

- ❑ To determine a route between any two routers, the algorithm simply finds the **shortest path** between them in the graph

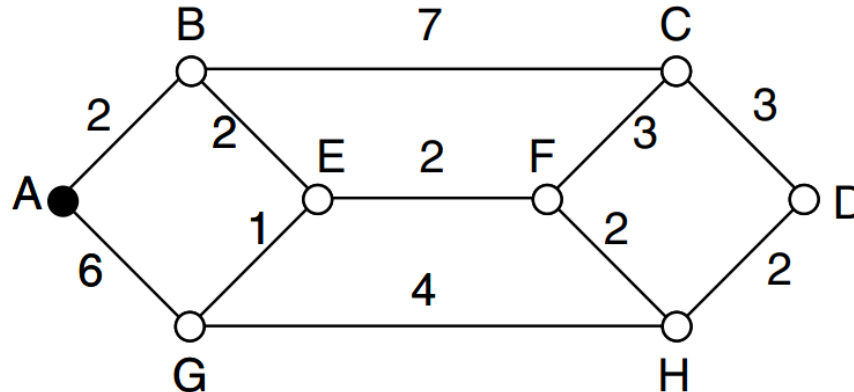
Shortest Path

- The notion of a **shortest path** requires some clarification:
- One way to measure path length is by counting the **number of hops**
 - ❖ using this metric, the paths ABC and ABE are considered equal
 - Another approach is to measure the **geographic distance** in kilometers, in which case ABC is clearly much longer than ABE



Shortest Path

- ❑ Many other metrics (beyond hop count and physical distance) can also be used
 - Each edge may be assigned a value representing the **average delay**
 - Then the shortest path is the **fastest path** rather than the path with the fewest edges or kilometers



Shortest Path

- ❑ In general, **edge labels** can be defined as **a function of factors** such as
 - Distance
 - Average traffic
 - Communication cost
 - Measured delay

- ❑ By modifying this weighting function, the algorithm can compute the “shortest” path based on a **single criterion** or a **combination of multiple criteria**.

Shortest Path: Dijkstra

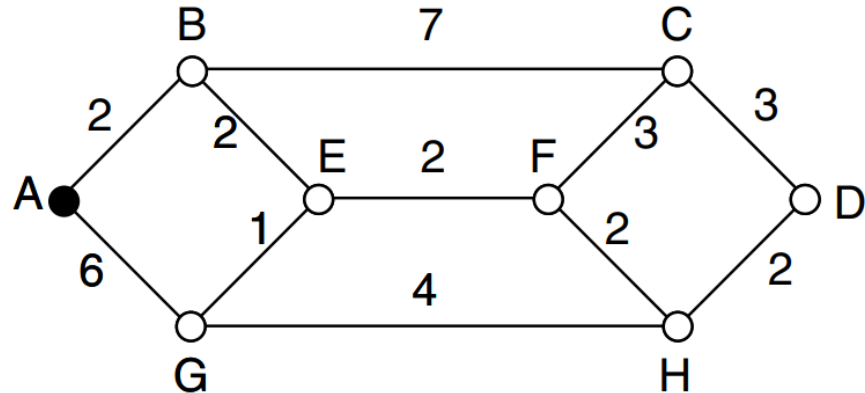
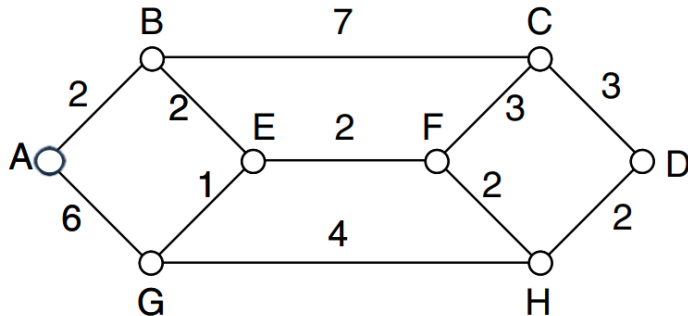
- ❑ Several algorithms exist for finding the shortest path between two nodes in a graph.
- ❑ One well-known approach is **Dijkstra** (1959), computes the shortest paths from **a single source node** to **all other nodes** in the network.
 - Each node is **labelled** with its distance from the source along the best-known path
 - These distances must be **non-negative**
 - They are based on **real metrics** such as **delay**
 - **Initially**, no paths are known, so all nodes are assigned a distance of **infinity**
 - A label can be either **tentative** or **permanent**. **Initially**, all labels are **tentative**
 - Once a label is confirmed to **represent the shortest possible path** from the source to a node, it becomes **permanent** and is not changed thereafter

Shortest Path: Dijkstra

- ❑ Initially, no paths are known, so all nodes are assigned a distance of infinity
- ❑ A label can be either tentative or permanent. Initially, all labels are tentative
- ❑ Once a label is confirmed to represent the shortest possible path from the source to a node
 - It becomes permanent and is not changed thereafter

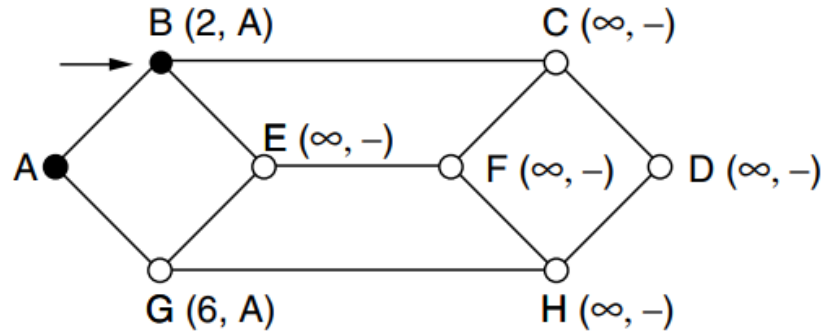
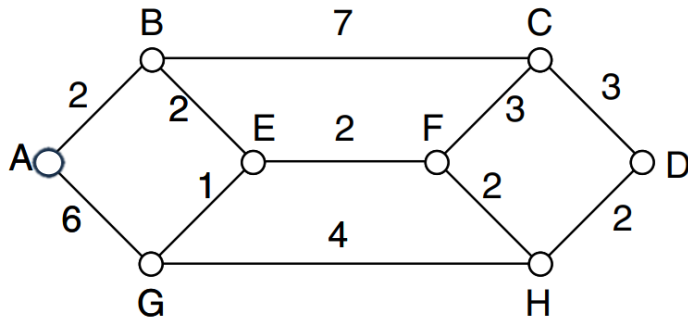
Shortest Path: Dijkstra

- To find the shortest path from A to D,
- we begin by marking node A as **permanent**, typically shown with a **filled circle**.
 - Then, we examine each node **directly connected** to A (the current working node) and **update their labels** with the **distance** from A.
 - Whenever a node is **reabeled**, it is also **tagged with the node from which it was reached**, allowing the **final path** to be reconstructed later.



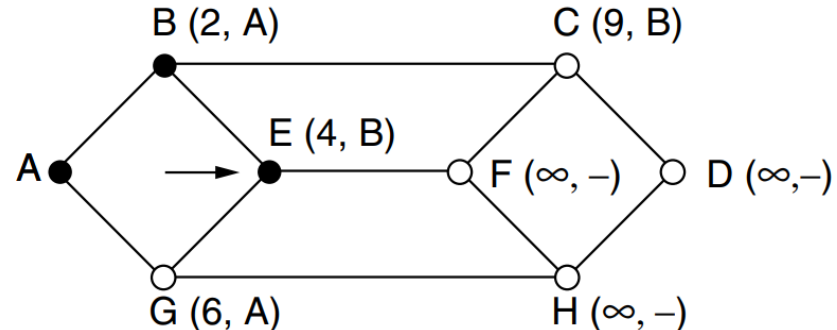
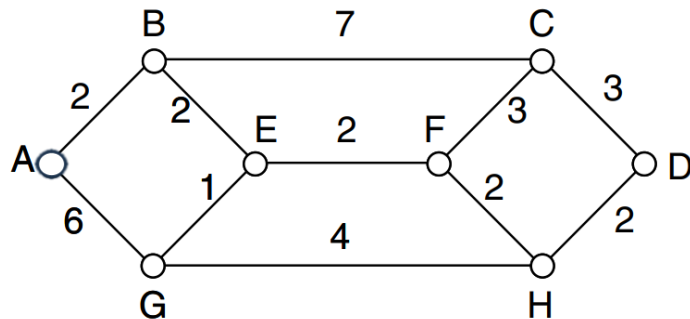
Shortest Path: Dijkstra

- After examining all nodes adjacent to A
 - we consider all **tentatively labeled** nodes in the graph and
 - select the one with the **smallest label**, making it **permanent**
 - ❖ So, Node B is selected as the **new working node**

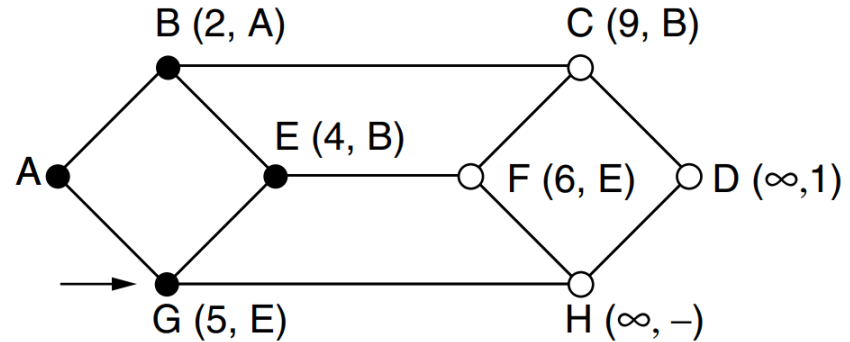
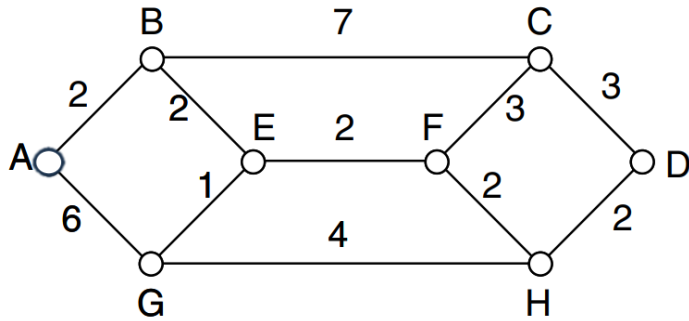


Shortest Path: Dijkstra

- ❑ We now **move to node B** and examine **all of its adjacent nodes**.
 - If the sum of **B's label and the distance from B to a neighboring node** is smaller than that node's **current label**
 - ❖ A shorter path has been found
 - The node is updated with the **new label (nodes E and C)**
- ❑ Then among all tentatively labeled nodes (C, E, G, F, D and H), the one with the **smallest label** is selected as the **new working node (E)**

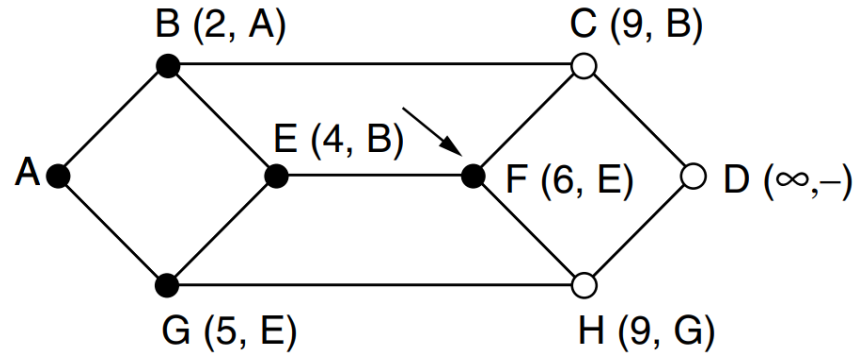
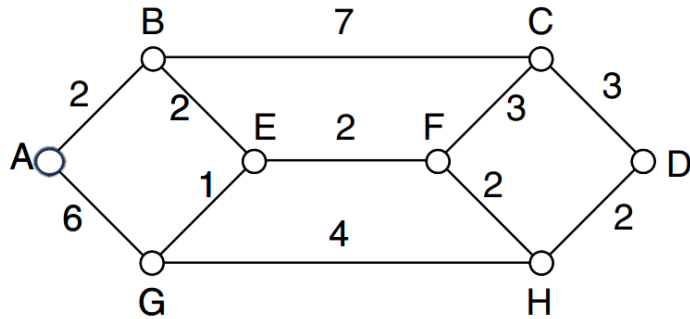


Shortest Path: Dijkstra



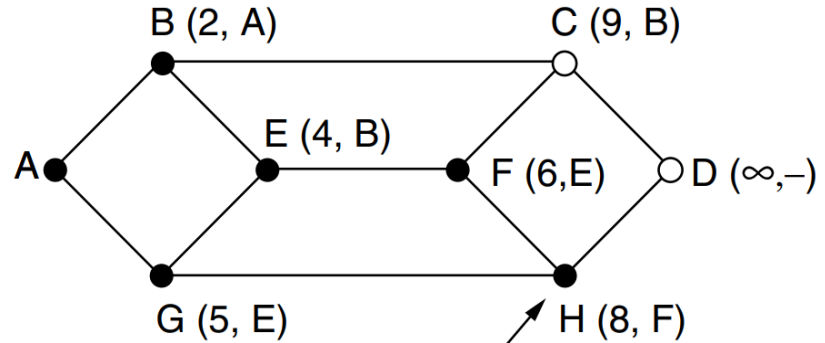
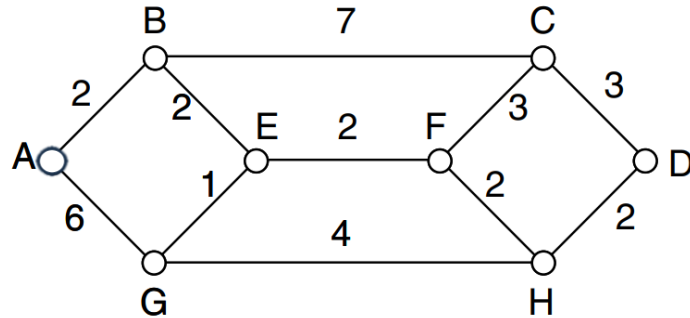
- ❑ The labels of nodes **F** and **G**, which are adjacent to **E** and **not yet permanent**, are updated.
 - Node **G** is then chosen as the next working node (the node with the smallest label among all tentatively labeled nodes is selected)

Shortest Path: Dijkstra



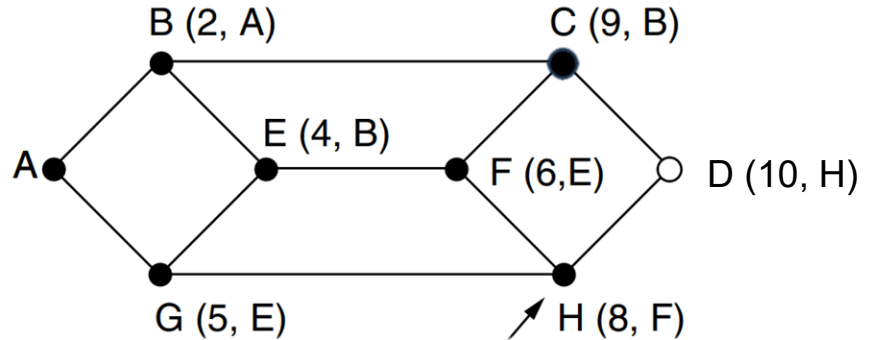
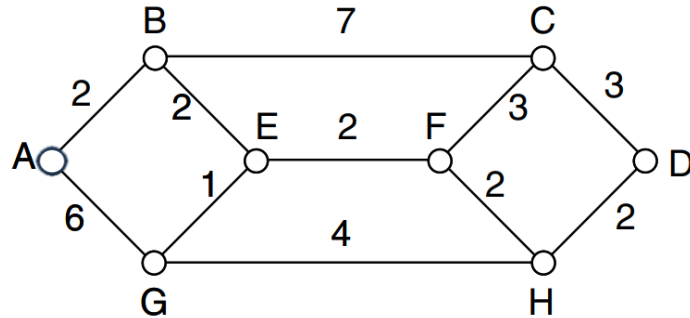
- ❑ The label of node H is updated.
 - Node **F** is then chosen as the next working node

Shortest Path: Dijkstra



- ❑ The label of node **H** is updated
 - Node **H** is then chosen as the next working node

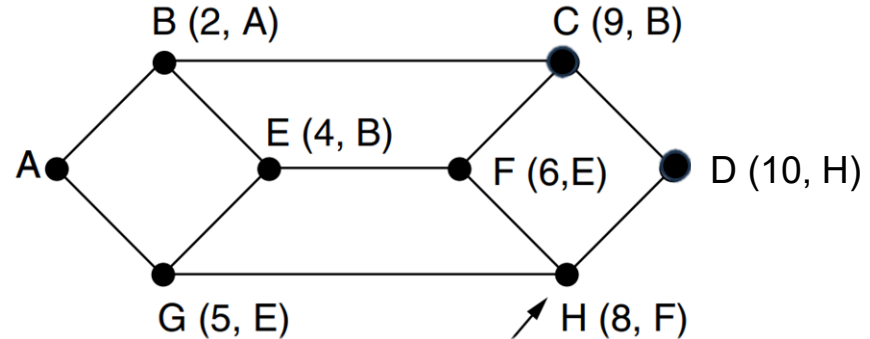
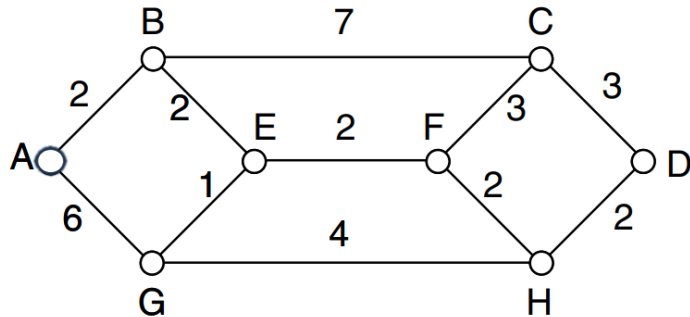
Shortest Path: Dijkstra



- ❑ The label of node **D** is updated
 - Node **C** is then chosen as the next working node

Shortest Path: Dijkstra

- ❑ Finally, node D is selected as the next working node
 - End of the algorithm



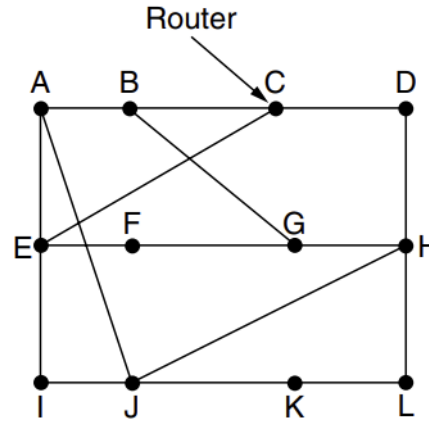
Distance Vector Routing

- ❑ Distance Vector Routing (**DVR**)
 - A **dynamic** routing algorithm
 - Sometimes called by other names (distributed Bellman-Ford routing)
 - Each router maintains a **table (vector)** that stores the shortest known distance to every destination and the corresponding outgoing link
 - Routers periodically **exchange** this information with their neighbors
 - ❖ Allowing them to update their tables
- ❑ Over time, all routers learn the best path to each destination

Distance Vector Routing

- ❑ The routing table is indexed by **destination** and contains **one entry** for each **router** in the network.
- ❑ Each entry includes two components:
 - The preferred **outgoing link** to reach that destination
 - An estimate of the **distance**
 - ❖ May be measured in hop count or using another routing metric
- ❑ Each router is **assumed** to know the distance to all of its **neighboring** routers

Distance Vector Routing



- Assume that J has estimated its distance to its neighbors, A (8), I (10), H (12), and K (6)

New estimated delay from J

To	A	I	H	K	Line	
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	–
K	24	22	22	0	6	K
L	29	33	9	9	15	K

JA delay is 8	JI delay is 10	JH delay is 12	JK delay is 6
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Vectors received from J's four neighbors

New routing table for J